



Dorset **Identification b.v.**

Weverij 26

7122 MS Aalten

T: +31(0)543 477119

F: +31(0)543 475355

E: id@dorset.nu

W: www.dorset.nu

Kvk 09067487

BTW nr. NL008.534.068.B01

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COMMAND SUMMARY LID650 & LID665 RS485 March 2015

This command summary explains the command set used by the LID650665N8xx decoder:

To control the decoders there are a couple of commands available.

All commands are given as **hexadecimal values** via a serial RS232/RS485 connection. When the decoder is delivered, its standard settings are as follows:

Data transfer:	19200 Baud
Data bits:	8 databits
Stop bit:	1
Parity:	None

The decoder can communicate at the following baud rates: 1200, 2400, 4800, 9600, 19200, 38400 and 57600.

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General string format

The strings send and received by the decoders have a certain format to ensure a secure data transfer on the communication-bus.

String format

<DLE><STX><Unit from><Unit to><Command><Data> <DLE><ETX><Checksum>

	String format	Remarks
<DLE><STX>	10h 02h	Start of frame indication
<Unit from> ¹	hexadecimal value between 00h-FEh	From address
<Unit to> ¹	hexadecimal value between 00h-FFh	To address
<Command> ¹	hexadecimal value between 00h-FFh	Value is specified by the command
<Data> ¹	hexadecimal value(s) between 00h-FFh	Values are specified by the command
<DLE><ETX>	10h 03h	End of frame indication
<Checksum>	hexadecimal value between 00h-FFh	XOR on all previous hexadecimal values

Recommendations

- The decoders cannot have unit number FFh. This value is used for broadcast commands to communicate with all decoders simultaneously.
- The decoders should not have unit number 00h. This is used by decoders which are started up for the first time. After first startup, the decoders' unitnumber should be altered to a value between 01h and FDh.
- The best unit number for the host controller (mostly a PC) is FEh.
- Be sure all decoders in the network have their own unique unitnumber. Double unitnumbers will cause errors in the communication.

Examples

The host (<Unit from>) = FEh sends commands to decoder (<Unit to>) = 01h

Start of frame	Unit from	Unit to	Command	Data	End of frame	Check sum	Final string to send
10h 02h	FEh	01h	54h	(none)	10h 03h	AAh	10 02 FE 01 54 10 03 AA
10h 02h	FEh	01h	6Ch	08h	10h 03h	9Ah	10 02 FE 01 6C 08 10 03 AA
10h 02h	FEh	01h	6Ch	10h	10h 03h	92h	10 02 FE 01 6C 10 10 10 03 92 ¹

Decoder number (<Unit from>) = 01h sends answers to the host (<Unit to>) = FEh

Start of frame	Unit from	Unit to	Command	Data	End of frame	Check sum	Final string to send
10h 02h	01h	FEh	54h	32h	10h 03h	AAh	10 02 01 FE 54 32 10 03 9A
10h 02h	01h	FEh	6Ch	08h	10h 03h	9Ah	10 02 01 FE 6C 08 10 03 AA
10h 02h	01h	FEh	6Ch	10h	10h 03h	92h	10 02 01 FE 6C 10 10 10 03 92 ¹

¹ when one of these values is 10h (<DLE>) an extra 10h (<DLE>) is inserted.

Command set overview

All bytes are send MSB first. The following commands are used to read and write the readers' settings. All these commands and responses are without:

<DLE><STX><Unit from><Unit to> and <DLE><ETX><Checksum>

Command name	Command + data	Response + data	Additional information
Log on	(empty message)	<i>vv vv vv</i>	<i>vv vv vv</i> = last 3 digits of the version number
Read memory status ('A')	41h	41h <i>aah aah nnh nnh</i>	<i>aa aa</i> = hexadecimal value of saved ID codes (0000h..FFFFh) <i>nn nn</i> = hexadecimal value of saved access codes
Read PWM pulse width to control a optional RC servo during idle ('A','1')	41h 31h	41h 31h <i>nnh</i>	<i>nn</i> = hexadecimal value of the pulse width. (00h-FFh)
Read PWM pulse width to control a optional RC servo during waittime ('A','2')	41h 32h	41h 31h <i>nnh</i>	<i>nn</i> = hexadecimal value of the pulse width. (00h-FFh)
Write PWM pulse width to control a optional RC servo during idle ('a','1')	61h 31h <i>nnh</i>	61h 31h <i>nnh</i>	<i>nn</i> = hexadecimal value of the pulse width. (00h-FFh). Value is relative and not a real angle in degrees
Write PWM pulse width to control a optional RC servo during waittime ('a','2')	61h 32h <i>nnh</i>	61h 32h <i>nnh</i>	<i>hh</i> = hexadecimal value of the pulse width. (00h-FFh) Value is relative and not a real angle in degrees
Read config byte ('C')	43h <i>nnh</i>	43h <i>nnh vvh</i>	<i>nn</i> = number of the config byte (31h..36h) <i>vv</i> = value of the config byte (00h..FFh) See page 9 for more information
Write config byte ('c')	63h <i>nnh vvh</i>	63h <i>nnh vvh</i>	<i>nn</i> = number of the config byte (31h..36h) <i>vv</i> = value of the config byte (00h..FFh) See page 9 for more information
Read date/time ('D')	44h	44h <i>aah bbh cch ddh eeh</i>	<i>aa</i> = seconds (0..59 BCD) <i>bb</i> = minutes (0..59 BCD) <i>cc</i> = hours (0..23 BCD) <i>dd</i> = years (0..3) & days (1..31) <i>ee</i> = day of week (0..6) & month (1..12) See page 14 for more information
Write date/time ('d')	64h <i>aah bbh cch ddh eeh</i>	64h <i>aah bbh cch ddh eeh</i>	<i>aa</i> = seconds (0..59 BCD) <i>bb</i> = minutes (0..59 BCD) <i>cc</i> = hours (0..23 BCD) <i>dd</i> = years (0..3) & days (1..31) <i>ee</i> = day of week (0..6) & month (1..12) See page 14 for more information

Erase all codes ('E')	45h	45h	Erases all saved ID codes from the readers memory. History codes and access codes
Erase history codes ('e')	65h	65h	Erases only the saved history codes
Give software trigger ('G')	47h	47h	The device accepted the software trigger and has started the scanning procedure
		bbh bbh bbh bbh bbh	bbbbbbbbbb = ID code. Length is depending on the transpondertype. Scanning procedure stopped
		00h 00h 00h 00h 00h	Trigger time has expired and no ID code was found.
Give command by device serial numer (<unit to> = FFh) ('j')	6Ah nnh nnh nnh nnh + command	command response	This command can be used to address 1 unit from more units with the same unitnumber
Reads devices software serial number ('J')	4Ah	4Ah nnh nnh nnh nnh	nnnnnnnn = Devices software serial number (00000000h..FFFFFFFh)
Enable relay for specified time ('k')	6Bh nnh tth	6Bh nnh tth	nn = relays who have to be set (31h..33h). 31h = relay 1 32h = relay 2 33h = relay 1 & 2. tt = (00h..FFh) time in 100ms
Read waittime after read ('L')	4Ch	4Ch tth	tt = waittime in 100 ms (00h..FFh)
Write waittime after read ('I')	6Ch tth	6Ch tth	tt = waittime in 100 ms (00h..FFh)
Read historical memory ('M')	4Dh	4Dh bbh bbh bbh bbh bbh ddh ddh ddh ddh	bbbbbbbbbb = ID code. Length is depending on the transpondertype. dddddddddd = Date/time stamp (when option is selected. See Configuration byte 2)
		4Dh	All the historical data was send
Read RAW data from the EEPROM ('M', 'R')	4Dh 52h	4dh 52h aah aah aah vvh vvh vvh vvh vvh vvh vvh vvh vvh vvh vvh vvh	aaaaaa = EEprom line start address (000000h..01FFF0h) vv..vv = 16 bytes RAW eeprom data
		4dh 52h	All EEprom lines were send
Read number readcycles ('R')	52h	52h nnnnh	nnnn = number of readcycles used for trigger and readdelay (0000h..FFFFh)
Write number of readcycles ('r')	72h nnh nnh	72h nnh nnh	nnnn = number of readcycles used for trigger and readdelay (0000h..FFFFh)

Switch RS485 receiver on decoder off ('S')	53h	53h	The RS485 receiver is switched off. This will prevent the reader to interrupt the readcycle while communication is on the RS485 bus. Receiver is switched on again after a disconnecting and connecting the power.
Read all access codes ('T')	54h	54h iih bbh bbh bbh bbh bbh	<i>ii</i> = Transponder type indication ² <i>bbbbbbbbbb</i> = ID code with access. Length is depending on the transponder type
		54h	All access codes are send
Write access code ('t')	54h iih bbh bbh bbh bbh bbh	54h iih bbh bbh bbh bbh bbh	<i>ii</i> = Transponder type indication ² <i>bbbbbbbbbb</i> = ID code. Length is depending on the transpondertype. After saving an access ID code, all the history will be cleared
Write unitnumber	75h nnh	75h nnh	<i>nn</i> = new unit number
Read firmware version number ('V')	56h	56h 4Ch 36h 35h 30h 36h 36h 35h 56h cch cch cch	<i>cccccc</i> = current firmware version
Read original firmware version number ('V','O')	56h 4Fh	56h 4Ch 36h 35h 30h 36h 36h 35h 56h cch cch cch	<i>cccccc</i> = original firmware version
Read bootloader version ('V','B')	56h 42h	56h 42h 4Ch 36h 35h 30h 36h 36h 35h 56h cch cch cch	<i>cccccc</i> = bootloader version
Write string to Trovan Flex transponder ('w')	77h bbh bbh bbh bbh bbh	77h bbh bbh bbh bbh bbh	<i>bbbbbbbbbb</i> = data to write to the Trovan Flex transponder. See page 18 for more information
Restart device ('x')	78h	78h	Restart device
Reset device to factory defaults ('X')	58h	58h	Reset to factory defaults

² see page 8, Transponder type indication

General commands

The following commands are used for general communication and give the decoder its general settings

Startup message

This message is sent by the decoder when it is powered up. This message is sent as a broadcast answer to unitnumber **FFh**.

When the unit is started up for the first time, the initial value of its unitnumber is **00h**. The message sent is called the cold startup message

Cold startup message sent by decoder:

10 02 00 FF 4B <last 3 digits version> **10 03** <checksum>

When the decoder already got its own unique unitnumber the warm startup message will be sent.

Warm startup message sent by decoder

10 02 <unit from> **FF 57** <last 3 digits version> **10 03** <checksum>

Broadcast

By sending a message to unit number **FFh**, all units will reply to this command and every unit will answer.

Log on

By sending an empty message to unit number **FFh**, all units will reply to this command and will give a log-on message including the version number of the unit.

*Example: (command from host **FEh**, to unit **FFh** (all units). Response from unit **0Ah**, with firmware version 800):*

String sent by host to decoder:

10 02 FE FF 10 03 00

Answer from decoder to host:

10 02 0A FE 38 30 30 10 03 CD

*Example: (command from host **FEh**, to unit **FFh** (all units). Response from unit **10h**):*

String sent by host to decoder:

10 02 FE FF 10 03 00

Answer from decoder to host:

10 02 10 10 FE 38 30 30 10 03 C7 (stuffed 10h at the location of the unitnumber)

Renumbering decoder

To give each decoder its own unique unitnumber this command can be used.

Command: **75** Renumber unit

Example: command from host FEh, to unit 00h, renumber to 38h):

String sent by host to decoder:
10 02 FE 00 75 38 10 03 B2

Answer from decoder to host:
10 02 38 FE 75 38 10 03 8A

Get Version

The version can be read by the following command.

Command:	56	Read the software version
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Example: (request from host **FEh**, version L650665N800 from unit **0Ah**)

String sent by host to decoder:

Answer from decoder to host:

```
10 02 0A FE 56 4C 36 35 30 36 36 35 4E 38 30 30 10 03 90
```

Reset

The decoder can be reset to its default values by the following command.

Command: **58** Reset the decoder

Example: (request from host FEh, reset unit 0Ah):

String sent by host to decoder:

Answer from decoder to host:
10 02 0A FE 58 10 03 AD

The decoder will reset immediately and give a cold start up message.

Cold startup message:

10 02 00 FF 4B <last 3 digits version> 10 03 <checksum>

LED/relaytime

default = 1h = 0,1 sec

The LED/relaytime is the time that the led LED resp. the relay is activated. This time can be read or written by the 'L' resp. 'l' command. The relay is activated as long as the LED lits. The configuration bytes decide if the LED and/or relay will be used.

Command: **4C** Read the current LED/relaytime
 Data: **hh**
hh = hexadecimal value of the number of readcycles (00h..FFh)

Example: (request from host **FEh**, response from unit **0Ah**, time = 5sec = **32h**):

String sent by host to decoder:

10 02 FE 0A 4C 10 03 B9

Answer from decoder to host:

10 02 0A FE 4C 32 10 03 8B

Command: **6C** Write a new LED/relaytime
 Data: **hh**
hh = hexadecimal value of the led/relaytime (00h..FFh)

Example: (request from host **FEh**, response from unit **0Ah**, time is set to 10 sec = **64h**):

String sent by host to decoder:

10 02 FE 0A 6C 64 10 03 FD

Answer from decoder to host:

10 02 0A FE 6C 64 10 03 FD

Configuration bytes

The configuration bytes handle the general settings of the decoder.

Configuration byte 1

Bitnumber	Function:	Default
7	Flash LED during a readcycle	1
6	Write the ID-codes to a LC-display. (only for L650, optional)	0
5	Turns the antenna on/off	1
4	Flash LED after an ID-code is read	1
3	Use buzzer after an ID-code is read	1
2	Use the software trigger to start a readcycle	0
1	Use hardware trigger input to start a readcycle	0
0	Save the read-only ID-codes in the historical memory	0

Configuration byte 2

Bitnumber	Function:	Default
7	Use relay after an ID-code is read (optional)	0
6	Save ID-code with date and time (optional)	0
5	Reserved	0
4	Reserved	0
3	Use the read-delay ¹⁾	0
2	Send only new ID-codes ¹⁾	0
1	Put the antenna in a fast mode (don't use with ANT610)	0
0	Use access control	0

¹⁾ When 'Read-delay' and 'send only new ID-codes' are used together the decoder will send only new ID-codes unless the same ID-code isn't read for the specified number of readcycles (see chapter 5 command N). Then the ID-code '0000000000' will be sent. If the **use relay** (config2.7) is activated the relay will be turned on while the ID number is present and will be turned off once the tag has left the antenna field.

If '**read-delay**' and '**send-only**' new ID codes are used in combination with the **use access control** function (config2.0), only ID numbers that are stored as keycard will activate the relay. Other ID numbers will not activate the relay

NOTE: '**use relay**' (config2.7) must be activated and the **LED/Relay time** (command L) must be > 1!

Configuration byte 3

Bitnumber	Function:	Default
7	Manual activation of relay	0
6	Reserved	0
5	Send the ID-code	1
4	Reserved	0
3	Overwrite history (1 = overwrite)	0
2	Baudrate selection for RS232 ²⁾	1
1	Baudrate selection for RS232 ²⁾	0
0	Baudrate selection for RS232 ²⁾	0

²⁾The baudrate can be selected as follows:

Baudrate	BIT2	BIT1	BIT0
1200 Bd	0	0	0
2400 Bd	0	0	1
4800 Bd	0	1	0
9600 Bd	0	1	1
19200 Bd	1	0	0

Configuration byte 4

Bitnumber	Function:	Default
7	Reserved	0
6	Reserved	0
5	Read Trovan Unique transponders	1
4	Read Trovan Flex transponders	0
3	Reserved	0
2	Reserved	0
1	Reserved	0
0	Reserved	0

Configuration byte 5

Bitnumber	Function:	Default
7	Use In/Out function (<i>only possible with special hardware!!</i>)	0
6	Save all movements (belongs to In/Out function)	0
5	Reserved	0
4	Copy relay status to output 2	0
3	Use Output 2 as PMW output (to control a RC servo)	0
2	Read 7 blocks Trovan Flex transponders	0
1	Read 5 blocks Trovan Flex transponders	0
0	Read 3 blocks Trovan Flex transponders	0

Configuration byte 6

Bitnumber	Function:	Default
7	Reserved	0
6	Reserved	0
5	Reserved	0
4	Reserved	0
3	Reserved	0
2	Read during waittime	0
1	Use output 3 to directly drive a IR LED (<i>only possible with special hardware</i>)	0
0	Bootmode flag (read only)	0

Configuration bytes 1 to 6 can be read or written by the following command.

Command: **43 nn** Read configuration byte **nn**
(31h....34h)

Example: (request from host **FEh**, response from unit **10h**, read configbyte 1, value is **68h**):

String sent by host to decoder:

10 02 FE 10 10 43 31 10 03 9D

Answer from decoder to host:

10 02 10 10 FE 43 31 68 10 03 E5

Command: **63 nn** Write configuration byte **nn**
(31h....34h)

Data: **nn** = hexadecimal value of the configuration byte

Example: (request from host **FEh**, to unit **0Ah**, write configuration byte 2, value = **00h**):

String sent by host to decoder:

10 02 FE 0A 63 32 00 10 03 A4

Answer from decoder to host:

10 02 0A FE 63 32 00 10 03 A4

Communication

After the decoder has read an ID-code, it can be sent immediately to an external device via RS232/485.

String sent by decoder:

10 02 0A FF *tt tt tt tt tt* 10 03 <Checksum>

Where *tt tt tt tt tt* is the Trovan® unique ID number

This function can be switched on or off by bit 5 of configbyte 3.

Transponder type indication

Type indication	Transponder type	Transponder length
0x20	Trovan Unique	10 characters
0x13	Trovan Flex 3 blocks	5 characters
0x15	Trovan Flex 5 blocks	10 characters
0x17	Trovan Flex 7 blocks	15 characters

Saving codes in the historical memory

In case a decoder is functioning stand-alone (not connected to another device) it is possible to save the ID-codes that are read in the decoders historical memory. This is done by setting bit 0 in configuration byte 1..

Number of saved codes

default 0000h

The number of saved codes in the historical memory can be read by the following command.

Command: **41** Read the number of saved codes

Data: *hh hh kk kk*
hh hh = hexadecimal value of the number of stored codes (0000h..FFFFh)
kk kk = hexadecimal value of the number of access codes (0000h..FFFFh)

Example: (request from host **FEh**, response from unit **0Ah**, data is **00 00 00 00**):

String sent by host to decoder:

10 02 **FE** 0A **41** 10 03 **B4**

Answer from decoder to host:

10 02 0A **FE** **41** 00 00 00 00 10 03 **B4**

The value of *hh hh* can be higher than the maximum number of codes that can be saved. This is to indicate how many codes are overwritten.

Deleting all saved codes

All saved codes in the historical memory can be deleted by the following command. The keycards (chapter 4) will still remain in the keycard table.

Command: **65** Delete all saved codes

Example: (request from host FEh, response from unit 0Ah):

String sent by host to decoder:

10 02 FE 0A 65 10 03 90

Answer from decoder to host:

10 02 0A FE 65 10 03 90

Reading saved codes

The saved codes, dates and times (optional) in the historical memory can be read by the following command. The date/time format is specified in the date/time command.

Command: **4D** Read all saved codes

Data: *Length dependant on transpondertype and date/time option*

Example: (request from host FEh, response from unit 0Ah, ID = tt tt tt tt tt, Date/time = aa bb cc dd ee):

String sent by host to decoder:

10 02 FE 0A 4D 10 03 B8

Answer from decoder to host when Trovan Unique was saved without date and time:

10 02 0A FE 4D tt tt tt tt tt 10 03 <Checksum>

Answer from decoder to host when Trovan Unique was saved with date and time:

10 02 0A FE 4D tt tt tt tt tt aa bb cc dd ee 10 03 <Checksum>

Answer from decoder (0Ah) to host (FEh) when all saved codes are sent:

10 02 0A FE 4D 10 03 B8

Date and time

The date and time can be read or set by the 'D' resp. 'd' command.

Command: **44**

Read date and time

Data: **aa bb cc dd ee**

Example: (request from unit **FEh**, response from unit **0Ah**):

String sent by host to decoder:

10 02 FE 0A 44 10 03 B8

Answer from decoder to host:

10 02 0A FE 44 aa bb cc dd ee 10 03 <Checksum>

aa = seconds (0..59),

bb = minutes (0..59),

cc = hours (0..23)

dd = years (0..3) and days (1..31)

MSB							LSB
7	6	5	4	3	2	1	0
Year (0 to 3 binary, where 0 is a leap year)		Ten days (0 to 3 binary)		Unit days BCD			

ee = Day of week (0..6) and months (1..12)

MSB							LSB
7	6	5	4	3	2	1	0
Day of week (0 to 6 binary)			Ten Months	Unit months BCD			

Command: **64**

Write date and time

Data: **aa bb cc dd ee**

aa = seconds (0..59),

bb = minutes (0..59),

cc = hours (0..23)

dd = years (0..3) and days (1..31)

MSB							LSB
7	6	5	4	3	2	1	0
Year (0 to 3 binary, where 0 is a leap year)		Ten days (0 to 3 binary)		Unit days BCD			

ee = Day of week (0..6) and months (1..12)

MSB							LSB
7	6	5	4	3	2	1	0
Day of week (0 to 6 binary)			Ten Months	Unit months BCD			

Example: (request from unit FEh, response from unit 0Ah):

String sent by host to decoder:

10 02 FE 0A 64 aa bb cc dd ee 10 03 <Checksum>

Answer from decoder to host:

10 02 0A FE 64 aa bb cc dd ee 10 03 <Checksum>

Using access control

The decoder can be used in small access control systems. When bit 0 in configuration byte 2 is set, approx. 1600 codes can be saved as keycards in the keycard table. If the code saving option is also selected (chapter 3) the maximum number of codes to be saved is decreased, because the decoder uses the same memory for both functions. When the saved codes in the historical memory are deleted (command 45) all the keycards will still be present in the keycard table.

Saving keycards

After bit 0 of configuration byte 2 is set, the keycards can be sent to the decoder using command 74. The first code of the keycard table is the goldcard, the second code is the silvercard. All other codes are keycards, which will all get access at any time. The gold- and silvercard can only be assigned via this command. The other keycards can also be added during the operation of the decoder by using the goldcard. If the gold- or silver-card is not used, it has to be assigned to all zeros.

Command: 74

Data: Dependant on which transpondertype is written

Example: (request from unit FEh, response from unit 0Ah):

String sent by host to decoder when writing a Trovan Unique keycard:

10 02 FE 0A 74 tt tt tt tt tt 10 03 <Checksum>

Answer from decoder to host when writing a Trovan Unique keycard:

10 02 0A FE 74 tt tt tt tt tt 10 03 <Checksum>

Note 1: When keycards are added to the keycard table, or when the 'use access control' bit (bit 0) of configuration bit 2 changes, all saved codes in the historical memory are lost. If these codes are important you must download these before you add one or more keycards!

Note 2: Never exceed the maximum number of 1600 keycards, cause the decoder will crash!

Reading keycards

All saved keycards can be read from the decoder using the **54** command.

Command: **54** Reads the keycards
Data: length dependant on transpondertype

Example: (request from unit **FEh**, response from unit **0Ah**):

String sent by host to decoder:

10 02 FE 0A 54 10 03 A1

Answer from decoder to host when a Trovan Unique is sent:

10 02 0A FE 54 tt tt tt tt tt 10 03 <Checksum>

Received string at the end of all data:

Answer from decoder to host:

10 02 0A FE 54 10 03 A1

Deleting all keycards

All saved codes including the gold- and silvercard are deleted by the **45** command.

Command: **45** Deletes all keycards

Example: (request from unit **FEh**, response from unit **0Ah**):

String sent by host to decoder:

10 02 FE 0A 45 10 03 B0

Answer from decoder to host:

10 02 0A FE 45 10 03 B0

Note 1: When the keycards are deleted from the keycard table, all saved codes in the historical memory are also lost. If these codes are important you must download these before you delete the keycards!

Other commands

Number of readcycles

default = 0002h

This value is used for the number of readcycles after a software- or hardware-trigger. It is also used by the readdelay option, the check in/out option and the In/Out function. The number of readcycles can be read or written by the 52 resp. 72 command. The total time of a readcycle depends on the connected antenna and the use of the fast reading mode.

Command: 52 Reads the current readdelay
Data: hhhh = hexadecimal value of the readdelay (0002h..FFFFh).

Example: (request from unit FEh, response from unit 0Ah, read default value 0002h):

String sent by host to decoder:

10 02 FE 0A 52 10 03 BB

Answer from decoder to host:

10 02 0A FE 52 00 02 10 03 BA

Command: 72 Writes a new readdelay
Data: hhhh = hexadecimal value of the readdelay (0000h..FFFFh).

Example: (request from unit FEh, response from unit 0Ah, write value 1A4Fh):

String sent by host to decoder:

10 02 FE 0A 72 1A 4F 10 03 D2

Answer from decoder to host:

10 02 0A FE 72 1A 4F 10 03 D2

RS232 trigger

When the software-trigger-bit in configuration byte 1 is set (bit 2), the decoder can start a number of readcycles by sending the 47 command.

Command: 47

Gives a software trigger

Example: (request from unit FEh, response from unit 0Ah):

String sent by host to decoder:

10 02 FE 0A 47 10 03 B2

Answer from decoder to host:

10 02 0A FE 47 10 03 B2 when trigger is received

When a transpondercode is read within the number of readcycles the code will be sent, otherwise 10 02 0A FE 00 00 00 00 00 10 03 F5 will be sent.

Write code to a Trovan Flex transponder (only for LID665)***Configuration of a Trovan Flex transponder***

Block 0 of the Trovan Flex transponder is the configuration block. Block 0 contains the following:

Bit number	Default value	Description	Trovan Flex 3 blocks config	Trovan Flex 5 blocks config	Trovan Flex 7 blocks config
0	0	Reserved	0	0	0
1	0	Reserved	0	0	0
2	0	Reserved	0	0	0
3	0	Reserved	0	0	0
4	0	Reserved	0	0	0
5	0	Reserved	0	0	0
6	0	Reserved	0	0	0
7	0	Reserved	0	0	0
8	0	Reserved	0	0	0
9	0	Reserved	0	0	0
10	0	Reserved	0	0	0
11	0	Reserved	0	0	0
12	0	Reserved	0	0	0
13	1	Reserved	1	1	1
14	0	Reserved	0	0	0
15	0	Reserved	0	0	0
16	0	Reserved	0	0	0
17	0	Reserved	0	0	0
18	1	Reserved	1	1	1
19	0	Reserved	0	0	0
20	0	Reserved	0	0	0
21	0	Reserved	0	0	0
22	0	Reserved	0	0	0
23	0	Reserved	0	0	0
24	X	Max block	0	1	1
25	X	Max block	1	0	1
26	X	Max block	1	1	1
27	0	Reserved	0	0	0
28	0	Reserved	0	0	0
29	0	Reserved	0	0	0
30	1	Reserved	1	1	1
31	0	Reserved	0	0	0

Dorset Identification b.v. reserves the right to change technical specifications of the product without prior notice.

To configure the Trovan Flex transponder, only Max block has to be changed. Max block is the amount of blocks the transponder will send when it enters the RF field from the antenna. Trovan Flex supports 3, 5 and 7 blocks (011, 101 and 111 binary).

Example:

Configure a Trovan Flex transponder as Trovan Flex 7 blocks:

the binary configuration string is:

0000 0000 0000 0100 0010 0000 1110 0010

We have to add 00100 as prefix and 010 as suffix:

00100 000000000000001000010000011100010 010

in hexadecimal this is: 2000210712, so the data will be 20 00 21 07 12

Command: **77** Writes a block in a read-write transponders

Data: **20 00 21 07 12**

String sent by host to decoder:

10 02 FE 0A 77 20 00 21 07 12 10 03 96

Received string:

10 02 0A FE 77 20 00 21 07 12 10 03 96

Write data to a Trovan Flex transponder

Read write transponders can be written by the **77** command. Read-write transponders contain 7 blocks but only blocks 2 till 7 can be used for data-storage.

A block is build up as follows:

0	D00	D01	P0
D02	D03	D04	P1
D05	D06	D07	P2
D08	D09	D10	P3
D11	D12	D13	P4
D14	D15	D16	P5
D17	D18	D19	P6
C0	C1	C2	P7

Dx – Databits x: 0..19

Py – Row Parity Bits y: 0..7

Cz – Column Parity Bits z: 0..2

All paritybits are odd parity. If all databits are 1, the block would be

0	1	1	1
1	1	1	0
1	1	1	0
1	1	1	0
1	1	1	0
1	1	1	0
1	1	1	0
1	0	0	0

This means a string of 01111110111011101110111011101000

The blocknumber is between 2 and 7 so in binary between 010 and 111

Blocknumber	2	3	4	5	6	7
Binary string to add	010	011	100	101	110	111

Add 00100 in front of the string and the binary string at the back 010

So writing to block 2 gives: 00100 01111110111011101110111011101000 010

In hexadecimal this will be 23F7777742 so the data will be 23 F7 77 77 42

Example: (request from unit FEh, response from unit 0Ah, write value 23 F7 77 77 42):
String sent by host to decoder:

10 02 FE 0A 77 23 F7 77 77 42 10 03 14

Received string: 10 02 0A FE 77 23 F7 77 77 42 10 03 14

A write example

How to write the text: "ROLAND" into the tag in block 2, 3 and 4.

First calculate the binary representation of the characters

Character	Hexadecimal value	Binary representation
R	52h	0101 0010
O	4Fh	0100 1111
L	4Ch	0100 1100
A	41h	0100 0001
N	4Eh	0100 1110
D	44h	0100 0100

Now write this binary information in the right order. {red is parity bits}

Block 2: ROL

0	0	1	0
0	1	0	0
0	1	0	0
0	1	0	0
0	1	1	1
1	1	0	1
1	0	0	0
1	0	1	1

Binary string is therefor: 0010 0001 0010 0010 0010 0011 1110 1100 0101 1010

The string to send is: 212223EC5A

Block 3: LAN

0	1	1	1
0	0	0	1
1	0	0	0
0	0	0	1
1	0	1	1
0	0	1	0
1	1	0	1
0	1	0	0

Binary string is therefor: 0010 0011 1000 1100 0000 1101 1001 0110 1010 0011

The string to send is: 238C0D96A3

Block 4: D

0	0	1	0
0	0	0	1
1	0	0	0
0	0	1	0
0	0	0	1
0	0	0	1
0	1	0	0
0	0	1	0

Binary string is therefor: 0010 0001 0000 1100 0001 0000 1000 1010 0001 0100

The string to send is: 210C108A14

This fits into a 5 blocks configuration, so we have to configure the transponder as Trovan Flex 5 blocks.